

Table 1: Direct comparison of original and PT2-expanded Siamese+MLP Raman+FT results on the original real-mixture dataset

Setup	Class space	Evaluation protocol	Precision	Recall	Micro-F1
Original <code>Siamese_MLP_6_FFT_Threshold</code> calibrated	12-class head	25% calibration / 75% held-out test, calibrated thresholds	0.970	0.980	0.970
PT2-expanded retrain, default scoring	17-class head	Full original real set, all 17 classes counted, fixed threshold 0.5	0.597	0.888	0.714
PT2-expanded retrain, notebook-style filtered	17-class head	Full original real set, report filtered to original supported classes only, fixed threshold 0.5	0.872	0.888	0.880
PT2-expanded retrain, recalibrated	17-class head	25% calibration / 75% held-out test on original real set, notebook-style filtered report, recalibrated 17-class thresholds	0.774	0.936	0.847

The original calibrated row is taken from `Siamese_MLP_6_FFT_Threshold.ipynb`.

“Notebook-style filtered” means the report only includes classes with nonzero support in the original real-mixture set, matching the notebook logic and ignoring false positives on absent classes.

The recalibrated PT2-expanded row uses newly fit 17-class thresholds from the new directory `pt2_augmented_existing_real_recalibrated`; its exact filtered micro metrics are precision 0.7738, recall 0.9356, micro-F1 0.8470.